Core-like satellites in a three-site Hubbard chain with positive U (FCI)

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Model. Three sites c, o, v (core, occupied, virtual), four electrons, on-site energies ϵ_i, hoppings t_{co}, t_{ov}, t_{cv} and only the on-site interaction $U_i \geq 0$:

$$H = \sum_{i,\sigma} \epsilon_i n_{i\sigma} + \sum_{i \neq j,\sigma} t_{ij}\, c^\dagger_{i\sigma} c_{j\sigma} + \sum_i U_i\, n_{i\uparrow} n_{i\downarrow}. \tag{1}$$

The ground state ($N = 4$) should be $\simeq c^2 o^2 v^0$. The removal spectrum is the exact (FCI) Lehmann sum over the $N = 3$ eigenstates m: a line at $\mathrm{IP}_m = E_m^{N-1} - E_0^N$ with weight $|\langle m|c_{i\sigma}|0\rangle|^2$ on site i (`scripts/fci_trimer.py`; Fock space of 64 states). Weights below are per spin and normalized to 1 for the c orbital. The *main line* is the strongest c line; a *satellite* is c weight at binding energy more than 1 above it. A point is accepted as a core only if $n_c > 1.9$, $n_v < 0.3$–0.4, the c lines carry less than 10% weight on o, v, all weighted IPs are positive, and the c line lies at least 3 above the o line.

Grid scan. 23 040 points ($\epsilon_c \in \{-14, -8\}$, $\epsilon_o = -1$, $\epsilon_v \in \{0.5, 2, 4\}$, $t_{co} \in [0.3, 2.5]$, $t_{ov} \in \{0.5, 1, 2\}$, $t_{cv} \in [0, 2]$, $U_i \in \{0, 3, 4, 6, 8\}$ only non-negative). With a deep core ($\epsilon_c + U_c < -3$) the best point is $\epsilon = (-8, -1, 0.5)$, $t = (0.6, 0.5, 1)$, $U = (4, 0, 6)$: main line 0.939 at IP = 4.37, one satellite of weight 0.0075 at IP = 13.4. Satellites stay below 1%.

Optimization. Differential evolution (6 seeds) maximizing the satellite weight, $U_i \in [0, 10]$, $t_{co}, t_{cv} \leq 0.5$, $t_{ov} \leq 4$, $\epsilon_c \in [-20, -3]$, $\epsilon_o \in [-5, 0]$, $\epsilon_v \in [0, 8]$, with the core criteria above. Five seeds give the same optimum, satellite weight 0.028 (the sixth, 0.017):

ϵ_c	ϵ_o	ϵ_v	t_{co}	t_{ov}	t_{cv}	U_c	U_o	U_v
−13.2	−3.55	0	0.50	0.15	0.50	9.95	3.13	0.05

with $n = (1.94, 1.77, 0.29)$. Its removal spectrum is in Fig. 1: the c weight is 0.392 and 0.541 at IP = 3.39 and 3.52 (a split main line), 0.010 at 4.77 and 0.017 at 16.8; the o line (weight 0.78 on o, 0.13 on v) is at 0.50. The upper parameters sit at the bounds ($t_{co}, t_{cv} = 0.5$, $U_c \approx 10$), so the maximum is set by the bounds, and the large U_c makes the c structure Hubbard-like rather than a shake-up.

Larger hopping. Letting t_{co}, t_{cv} go up to 3 raises the weight of the second c line to 0.24 ($\epsilon = (-10.2, -3.95, 4.85)$, $t = (2.15, 1.91, 2.94)$, $U = (0.42, 4.56, 9.87)$; main line 0.73 at IP = 10.82, second line 0.22 at 10.02). This line lies 0.8 *below* the main line and the parameters are at the bounds with strong c–o–v hybridization, so it is a level splitting, not a shake-up satellite.

Conclusion. With on-site $U_i \geq 0$, hoppings and local energies only, a core hole reaches o and v only through hopping, so the satellites are weak.

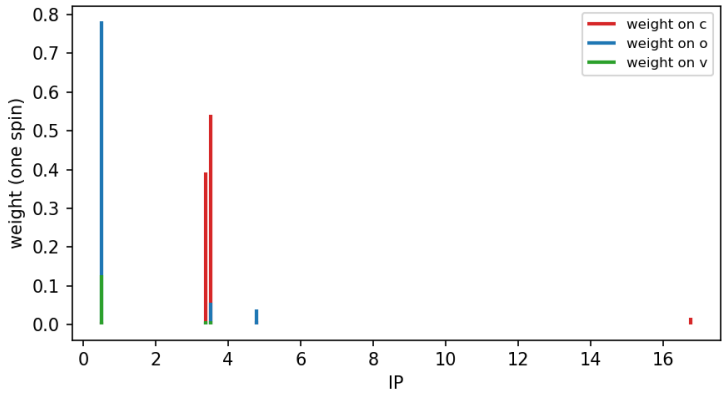


Figure 1: Removal spectrum (one spin) of the best $U \geq 0$ point, weights on c, o, v.

- *Largest satellite weight with a genuine core (shake-up side):* 0.028 of the c weight, at $\epsilon = (-13.2, -3.55, 0)$, $t_{co} = 0.50$, $t_{ov} = 0.15$, $t_{cv} = 0.50$, $U = (9.95, 3.13, 0.05)$. Its nature: 0.017 is a far satellite at IP = 16.8 (about 13 above the main line, a true shake-up-like line) and 0.010 is a c–o mixed line at 4.77 (1.3 above the main line; this line has 0.039 weight on o). The main line itself is split ($0.39 + 0.54$ at 3.39, 3.52; two lines 0.13 apart), with $U_c \approx 10$ and $t_{co}, t_{cv} = 0.5$ at the optimizer bounds.
- *Largest weight overall:* 0.24, at $\epsilon = (-10.2, -3.95, 4.85)$, $t = (2.15, 1.91, 2.94)$, $U = (0.42, 4.56, 9.87)$. Its nature: a level splitting of c by strong c–o–v hybridization (second line 0.8 below the main line), not a shake-up.
- *Cleanest deep core (grid):* $\epsilon = (-8, -1, 0.5)$, $t = (0.6, 0.5, 1)$, $U = (4, 0, 6)$: qp 0.939, satellite 0.0075 at IP = 13.4.